

The Möbius-Screw Electron: Framed Unknot Geometry for $g = 2$ and a Capacitance Model of α

Companion note to the Flag Condensate phase-slip programme

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July 22, 2026 (revised July 26, 2026: post-falsification remark added)

Abstract

We present a geometric model of the electron as a *framed unknot*: the centerline of a $(2, 1)$ -torus embedding on a torus of major radius R and minor radius a . Reparameterisation shows that the knot type is trivial (an unknot). The torus framing has self-linking magnitude $|Sl| = pq = 2$ and a spinorial lift. The earlier identification of this magnitude with $g = 2$ is withdrawn: parity does not determine a magnetic moment. A separate capacitance estimate on the double-cover (Möbius) annulus, with self-stress matched to $m_e c^2$, returns a numerical value $\alpha^{-1} \approx 137.036$ under stated approximations. The phase-slip / Bogoliubov channel that links interior standing waves to exterior travelling waves is identified with the companion flag-condensate nuclear-decay note. All statements are model claims with explicit scope; this note does not claim a derivation of QED or a proof of the measured g -factor beyond the stated geometric identification.

Publication status (24 September 2026). This version includes a claim-specific amendment at the end of the main text. That amendment governs earlier physical interpretations and distinguishes retained calculations from withdrawn or unresolved inferences.

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1 Scope and claim discipline

This note is a geometric *design* for an electron model inside the flag-condensate programme. It is not a theorem of QED, not a measurement, and not a uniqueness claim.

- **In scope:** centerline geometry; torus-knot winding numbers; framing and self-linking, with the proposed origin of $g = 2$ withdrawn; a capacitance estimate for α^{-1} ; the link to the phase-slip transfer-matrix picture.
- **Out of scope:** radiative corrections to $g - 2$; full QED renormalisation; a first-principles derivation of R and a from the Standard Model Lagrangian; experimental claims beyond the numerical capacitance estimate under stated approximations.

2 Centerline and reparameterisation

2.1 Parametric form

Let $\varphi \in [0, 4\pi)$. The centerline is

$$\mathbf{r}(\varphi) = \begin{pmatrix} (R + a \cos(\varphi/2)) \cos \varphi \\ (R + a \cos(\varphi/2)) \sin \varphi \\ a \sin(\varphi/2) \end{pmatrix}. \quad (1)$$

2.2 Reparameterisation: $(2, 1)$ -torus embedding

Set $t = \varphi/2$. As φ runs over $[0, 4\pi)$, t runs over $[0, 2\pi)$, and

$$\mathbf{r}(t) = \begin{pmatrix} (R + a \cos t) \cos(2t) \\ (R + a \cos t) \sin(2t) \\ a \sin t \end{pmatrix}, \quad t \in [0, 2\pi). \quad (2)$$

This is a torus curve on a torus of major radius R and minor radius a . The longitude angle is $2t$ and the latitude angle is t . The winding numbers are therefore

$$p = 2 \quad (\text{longitude}), \quad q = 1 \quad (\text{latitude}). \quad (3)$$

The knot type is a $(2, 1)$ -torus knot.

Remark 1 (Correction of winding order). *An earlier draft labelled the object a “ $(1, 2)$ -torus knot.” Under the standard convention that p is the longitudinal winding and q the meridional winding, the reparameterisation (2) identifies a $(2, 1)$ embedding. The distinction matters for reading the self-linking number as $p \cdot q$.*

2.3 Knot type: framed unknot

Because one winding number equals 1, the knot itself is trivial — an unknot. The spin-1/2 content in this model does not arise from a non-trivial knot (e.g. a trefoil) but from the *framing* of the unknot induced by the torus embedding.

Definition 1 (Model electron). *In this note, the electron is modelled as a **framed unknot**: a simple closed loop carrying two full twists of framing, equivalently a $(2, 1)$ -torus embedding with self-linking number 2.*

3 Self-linking and $g = 2$

3.1 Călugăreanu identity

For a closed framed curve,

$$Sl = Tw + Wr, \quad (4)$$

where Sl is the self-linking number, Tw the total twist, and Wr the writhe [1, 2, 3].

3.2 Torus framing

A (p, q) -torus knot inherits a natural framing from the torus embedding. Under that framing the self-linking number is

$$Sl = p \cdot q = 2 \cdot 1 = 2. \quad (5)$$

3.3 Geometric identification with $g = 2$

In the Dirac theory the free-electron gyromagnetic ratio is $g = 2$ at tree level. In the present model we identify

$$g \longleftrightarrow Sl = 2 \quad (6)$$

as a topological bookkeeping statement: the framed unknot stores two units of linking, matching the 4π (two-turn) return of a spin-1/2 frame. This is a model identification, not a derivation of the QED vertex form factor.

Proposition 1 (Model claim). *Within the framed-unknot electron model defined by (2) and (5), the geometric self-linking number equals 2, and is identified with the tree-level value $g = 2$.*

Remark 2 (Superseded: the identification (6) is falsified (T4)). *The companion note [5] computes the $SU(2)$ lift of this model's frame loop and finds that the spin content of the self-linking number is carried by its parity alone: for torus curves the holonomy is $\sigma = (-1)^{p+q} = (-1)^{Sl+1}$, so $Sl = 2$ and $Sl = 0$ lie in the same class of $\pi_1(SO(3)) \cong \mathbb{Z}/2$. An untwisted round circle is therefore spinorial in exactly the same sense as the framed unknot here, and the value 2 carries no spin information that 0 does not. The identification (6) accordingly loses its geometric motivation and is recorded as falsified in the Elimination Ledger.*

The geometry of §2 and the computation $|Sl| = pq = 2$ are unaffected — both are confirmed numerically in [5] — and so is the conclusion that this shape is spinorial. What changes is the source: the double cover comes from the odd meridian winding $q = 1$, i.e. the $\varphi/2$ half-angle in (1), not from the product pq .

3.4 Spin radius

Spin angular momentum of magnitude $\hbar/2$ for a circulating mass m_e at speed c on a circle of radius R fixes, in the usual order-of-magnitude estimate,

$$R = \frac{\hbar}{2m_e c}. \quad (7)$$

Equation (7) is an input scale for the geometric model, not an independent prediction.

4 Capacitance estimate for α

4.1 Double-cover annulus

Treat the Möbius / double-cover geometry as an effective annular conductor of characteristic radius R and thickness scale a . A standard approximate capacitance for a thin ring / annulus form is

$$C \approx \frac{2\pi\epsilon_0 R}{\ln(8R/a) + 1}. \quad (8)$$

(The additive constant in the denominator is scheme-dependent at this order; the logarithm dominates for $R \gg a$.)

4.2 Geometric fidelity of the capacitance model

Equation (8) is an *annulus / thin-ring* approximation to the electrostatic energy of the Möbius ribbon. The actual double-cover geometry is a twisted strip (or its immersion as a one-sided surface), not a flat annulus. Two consequences follow:

1. The logarithmic cutoff $\ln(8R/a)$ encodes an effective aspect ratio; a different conformal modulus for the true ribbon will shift the constant term and, more weakly, the argument of the logarithm.
2. The factor $e/2$ per sheet is a double-cover bookkeeping choice; alternative charge assignments change the prefactor of E_{cap} at the same order as a change of conformal modulus.

A natural revision path is to replace (8) by the capacitance of the ribbon domain under a conformal map to a standard annulus (or by a boundary-integral computation on the immersed strip), then re-solve (9). That revision is carried out as an as-implemented numerical sequel in [6] (annulus vs conformal-modulus vs Möbius BIE collocation, with aspect-ratio scan and machine-readable results). Under those revisions, moderate geometric aspect ratios yield $\alpha^{-1} = \mathcal{O}(1)$ rather than ≈ 137 ; the figure $\alpha^{-1} \approx 137.036$ remains a numerical output of the annulus model at a tuned cutoff, not a geometry-unique invariant of the framed unknot.

4.3 Self-stress matching

Assign charge $e/2$ to each sheet of the double cover and match the electrostatic self-energy to the rest energy:

$$E_{\text{cap}} = \frac{(e/2)^2}{2C} = m_e c^2. \quad (9)$$

Combining (8)–(9) with (7) and solving for the fine-structure combination $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ yields, under the stated approximations,

$$\alpha^{-1} \approx 137.036. \quad (10)$$

Remark 3 (Status of (10)). *The estimate is cutoff- and charge-split-sensitive; see §4.2. It is reported as an as-implemented estimate, not as a uniqueness theorem for α .*

5 Phase-slip unification

The companion note on nuclear decay in the flag condensate [4] treats decay as a spectral bifurcation: a confined standing-wave phase lock in the throat becomes an exterior travelling wave through Bogoliubov mode mixing across a geometric phase slip, with

$$\left| \frac{\beta}{\alpha} \right|^2 = e^{-2W} \quad (11)$$

(Gamow / WKB). The same transfer-matrix structure recovers Hawking temperature on a Schwarzschild horizon in that note.

In the present electron model, the “throat” is the high-torsion region of the framed unknot (the region of strongest expansion/contraction of the screw). The phase-slip channel is the same formal object: interior standing structure $\leftrightarrow$ exterior travelling structure. No new numerical nuclear or Hawking fits are claimed here; the identification is structural.

6 Summary of formal content

Object	Status in this note	Scope
$(2, 1)$ centerline (2)	geometric definition	model
unknot + framing $Sl = 2$	topological reading	confirmed, $ Sl = 2$
$g \leftrightarrow 2$	falsified (T4)	see Remark 2
frame loop spinorial	$\sigma = -1$ in $\pi_1(SO(3))$	confirmed, from $q = 1$
$R = \hbar/(2m_e c)$	scale input	order-of-magnitude
C estimate (8)	approximate formula	capacitance model
$\alpha^{-1} \approx 137.036$	numerical estimate	under (9)
phase-slip / Bogoliubov	structural link	see [4]

Acknowledgments

This note consolidates the framed-unknot correction (twist, not knotting) with the capacitance and phase-slip sketches of the Flag Condensate Collaboration draft.

Amendment: scope and evidence, 24 September 2026

The framed-curve geometry survives, but the inference from self-linking to $g = 2$ is withdrawn. The spin-cover class is a parity datum, whereas the gyromagnetic ratio depends on charge/current and mass distributions. For uniform charge-to-mass ratio on the same classical closed curve, the conventional ratio is $g = 1$. The capacitor model also retains a cutoff and charge split. For $C = 2\pi\epsilon_0 R/L$, charge $e/2$ and $R = \hbar/(2mc)$, energy matching gives $\alpha^{-1} = L/2$, not L . These findings preclude treating either estimate as a parameter-free particle derivation.

For the complete-action and microscopic-carrier audit, see the companion papers *Complete Scalar Actions and Identifiability Boundaries in the ACS Framework* and *Condensate Binding, Charge Protection, and the Boundary of Particle Identification* in this repository's `papers/updates/` directory. The dated publication record separates fresh portable checks from preserved archival results; it does not claim a complete rerun of every historical experiment or a proof of RH.

References

- [1] G. Călugăreanu, *Sur les classes d'isotopie des noeuds tridimensionnels et leurs invariants*, Czechoslovak Mathematical Journal **11** (1961), 588–625.
- [2] J. H. White, *Self-linking and the Gauss integral in higher dimensions*, American Journal of Mathematics **91** (1969), 693–728.
- [3] F. B. Fuller, *The writhing number of a space curve*, Proceedings of the National Academy of Sciences **68** (1971), 815–819.
- [4] Flag Condensate Collaboration, *Nuclear Decay as a Spectral Bifurcation of the Flag Condensate*, preprint (July 2026), `papers/notes/Flag_Condensate_Nuclear_Decay.tex`.
- [5] Flag Condensate Collaboration, *The Framing Transformer: Where the Spin-1/2 Content of the Möbius-Screw Actually Lives*, preprint (July 2026), `papers/notes/Framing_Transformer_Spin_Parity.tex`.
- [6] Flag Condensate Collaboration, *Möbius-Ribbon Capacitance: Annulus, Conformal Modulus, and BIE Revisions of the α Estimate*, preprint (July 2026), `papers/notes/Mobius_Ribbon_Capacitance.tex`.

A Interpretive framing: tornado metaphor

This appendix is interpretive only. It does not add formal claims to Sections 2–5.

Think of the model electron not as a static ribbon but as a self-sustaining vortex in a condensate: an Archimedean screw as rotating updraft, narrowing from a boundary field toward a projected interior. The throat is the high-torsion neck; the swirling debris is the phase-locked visible signature of the invisible twist. The quiet axis — the eye — is read as the coherence fulcrum of the structure, not as a literal empty void.

The tornado image is useful because it emphasises what the mathematics already says: power lives in framing (stored torsional stress, self-linking), not in knotting. The electron, in this metaphor, is a localised high-amplitude closed feedback loop — a framed unknot that persists because its twist is topologically locked.